

Literature Status: Universal Differentiability Sets in Laakso Space

Automatic Math Discovery Run `fa.banach.001`

July 3, 2026

Status

This is a lightweight literature-status packet. The source paper [1] lists the existence of universal differentiability sets in Laakso spaces as still open. A later paper [2] explicitly answers that question affirmatively.

Source question

In the final section, “Further developments,” Capolli’s overview [1] states that the existence of UDS in Laakso spaces is still open. It explains that the earlier paper of Capolli–Pinamonti–Speight proved only that the classical approach does not work in the Laakso space, leaving the existence of UDS open.

Supporting answer

Eriksson-Bique, Pinamonti, and Speight [2] explicitly identify the same gap in their introduction: previous work left open whether measure-zero UDS exist in Laakso space, and their paper answers this question by a different method. Their Theorem 2 states that there is a Borel set $N \subset F$ with

$$\mathcal{H}^Q(N) = 0$$

such that every Lipschitz map $f : F \rightarrow \mathbb{R}$ is differentiable at some point of N . Thus the Laakso-space UDS existence question from [1] is answered affirmatively in the later literature.

Scope limitations

This identification answers only the UDS existence item for the standard Laakso space F . It does not settle the other broad further-development questions in [1], such as Laakso-type spaces based on other self-similar sets, non-Abelian Carnot models, or general inverse-limit spaces.

References

- [1] M. Capolli, *An Overview on Laakso Spaces*, arXiv:2211.05681.
- [2] S. Eriksson-Bique, A. Pinamonti, and G. Speight, *Universal Differentiability Sets in Laakso Space*, arXiv:2406.07660.